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The impact of electricity price on inflation in the Czech Republic: An ARDL analysis

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Declaration

I declare that I am the sole author of the thesis entitled *The impact of electricity price on inflation in the Czech Republic: An ARDL analysis* I duly marked out all quotations. The used literature and sources are stated in the attached list of references.

May 2024, Prague

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Student's signature

Acknowledgements

I would like to sincerely thank my mother, who is always here for me, and my supervisor Ing. Mgr. Lukáš Frýd Ph.D. for his help patience, and consultations during this period.

Abstrakt

V této práci jsem provedla analýzu vztahu CPI a tržních cen elektřiny a jejích změn v letech 2015 až 2019 v České republice. Cílem této bakalářské práce je zjistit, jak cenová hladina byla ovlivněna cenami elektřiny pomocí ARDL modelu. Tato metodologie je vhodná pro práci s nestacionárními daty a umožňuje odhalit dlouhodobou i krátkodobou dynamiku mezi proměnnými. Výsledky ukázaly, že mezi těmito proměnnými existují oba typy vztahů.

Klíčová slova

Index spotřebitelských cen, ARDL, cena elektřiny

Abstract

In this thesis, I conducted an analysis of CPI and whole market electricity prices relationship and its changes between 2015 and 2019 in the Czech Republic. The objective of this bachelor's thesis is to investigate how price level has been influenced by electricity by creating an ARDL model. This methodology is well-suited for handling non-stationary data and allows to reveal both long-term and short-term dynamics between variables. Results showed that both types of relationships are present between these variables.

Keywords

CPI, ARDL, electricity prices

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Seznam použitých zkratk

ARDL Autoregressive Distributed Lag

ADF Augmented Dickey–Fuller test

ACF Autocorrelation function

PACF Partial Autocorrelation Function

CPI Consumer price index

Introduction

Inflation has become a frequent topic of conversation in recent years. This heightened awareness can be attributed in large part to significant inflationary pressures resulting from global events such as the COVID-19 pandemic and the subsequent Russian-Ukrainian war. The imposition of sanctions in response to the conflict has had a significant impact on energy markets, worsening inflationary pressures.

Traditionally, research on the impact of energy price shocks on macroeconomic variables has mainly focused on crude oil prices (1) (2) (3) (4). While these studies provide valuable insights, they often focus on generalized energy markets. Particularly, there is a lack of research focusing on the isolated primary impact of electricity prices.

Since electricity is a non-storable commodity, supply and demand must coincide at every moment. Any imbalances between them have substantial cost implications such as the rescheduling of the supply system and grid congestion which ultimately affect electricity prices. Additionally, the unpredictable nature of renewable energy generation can lead to unexpected fluctuations in day-ahead and real-time electricity prices. Unforeseen variations in renewable energy sources can also cause unpreparedness and scheduling issues, further contributing to price volatility.

Energy production costs encompass the expenses associated with producing, transporting, and delivering electricity to customers at specific times and locations. Examples of production costs are those related to the fixed and variable costs of power plants, grid costs, and the cost of renewables integration. Transmission and distribution (T&D) costs, arising from the power lines and grid infrastructure, are another component of the overall cost structure. Both energy production and T&D costs are directly paid to suppliers by local retailers.

The production cost channel is one of the primary channels through which electricity prices can affect core inflation. When electricity prices increase, it leads to an increase in the production costs of firms that rely heavily on electricity as an input. This increase in production costs is then passed on to consumers in the form of higher prices, contributing to higher core inflation. This channel is particularly important for industries that are highly dependent on electricity, such as manufacturing and mining.

Another possible way the pass-through effect can also occur is through the wage channel. When electricity prices increase, it can lead to higher production costs, which can then lead to higher wages as firms try to maintain their profit margins. This increase in wages can then lead to higher prices and higher core inflation. This channel is particularly important for industries that are heavily dependent on labor, such as services and construction.

Additionally, electricity prices can affect core inflation through the consumption channel. When electricity prices increase, it reduces the purchasing power of consumers, leading to a decrease in their consumption patterns. This decrease in consumption can lead to lower

aggregate demand, which can then lead to higher prices and higher core inflation. This channel is particularly important for low-income households, who spend a larger proportion of their income on electricity and other essential goods.

According to statistics, in 2019 26,3% of household expenditure is housing, and a fifth of that is electricity, gas, and other fuel (5), the increase in the price of electricity represents a significant burden on family budgets. Given this substantial share and energy dynamics, this research aims to investigate whether or how electricity prices impact the Consumer Price Index (CPI) in the Czech Republic.

Constructed within the framework of the empirical investigation Autoregressive Distributed lag (hereafter referred to as ARDL) model will utilize data from January 2015 to December 2019 (60 monthly observations). This timeframe excludes the years 2020-2023 due to the significant exogenous shocks described above that could bias the model estimates. Unfortunately, due to the limited data sample, it is not possible to implement a segmented approach where the data is split at structural breakpoints.

The results of the empirical analysis, presented in the form of different model variations, will be detailed in the final part of this work, offering insights into the relationship between electricity prices and inflation in the Czech Republic.

1. Literature review

The pass-through mechanism in the energy sector serves as a critical link between energy price fluctuations and broader economic indicators.

Cost pass-through arises when a business changes the prices of the products or services it supplies following a change in its costs.⁽⁶⁾ This phenomenon is particularly relevant for commercial energies, like coal, oil, gas, and electricity, which are traded wholly or almost entirely in the marketplaces. ⁽⁷⁾ Their market price can lead to cost pressures on businesses and eventually cause price adjustments for goods and services.

The tendency to return to the topic of transition changes in the energy sector to the core inflation resumes after major economic shocks, such as the energy crisis of the mid-70s^{(8),(2)}, the outbreak of the Iraq War, or the global financial crisis in 2008. Since the beginning of this Russian-Ukrainian war in Europe, the relevance of energy costing is high again.

The specificity and focus on primary products have historically inclined researchers towards considering the transmission mechanism primarily through the lens of oil and by-products during its production.

While Mork, K.A.; Hall, R.E.⁽⁸⁾ identified a pronounced effect of energy price increases on inflation, De Gregorio et al. ⁽¹⁾ offer evidence of a reduced pass-through of oil prices to domestic inflation. In light of analysis, this study reveals a more significant decline in the pass-through effect in advanced economies. It associates it with a reduction in oil intensity and the extent of exchange rate transmission.

Habermeier and colleagues ⁽⁹⁾ analyzed panel data from 50 countries spanning 2007-2008 years and discovered that monetary policy has important implications for the spread of oil and food shocks. Their results show that countries that adopt an inflation-targeting framework and have higher levels of central bank independence tend to have lower transmission rates. Additionally, Álvarez et al. ⁽³⁾ have observed changing dynamics in the euro area, where the direct impact of high oil prices on inflation is increasing as households spend more on refined petroleum products. However, the subsequent secondary and indirect effects decreased over time.

Nicholas Apergis ⁽¹⁰⁾ in his work dedicated to analyzing the causal relationship of CPI components in Greece also claimed that the primary price movements are transmitted from the energy price indices, i.e. the electricity price index, the energy price index, and the fuels and gas price index, while secondary effects arising from the food and vegetables price index as well as from the services price index.

Another cluster of associated literature studies the time-varying aspects of oil price shocks. The literature emphasizes the underlying drivers of oil price fluctuations are important in shaping the macroeconomic impact. As illustrated by Kilian⁽⁴⁾, Peersman and Van

Robays(11), and Baumeister and Peersman (12) the origin of oil price movements, whether stemming from a positive demand shock or a negative supply shock, leads to distinct impacts on both real GDP and inflation.

For the most part, recent studies find limited utility in using international crude oil price shocks as explanatory factors mainly because crude oil is not used directly by consumers and is at least not a direct input to companies outside the refining industry.

The electricity cost pass-through effect refers to the phenomenon in which changes in the price of electricity are reflected in the prices of goods and services. While price increases for businesses seem likely, the extent to which these costs will be passed through depends on several factors.

- **Competition.** A meta-analysis by Borenstein et (13) examined electricity price pass-through in various industries across multiple countries. They found that competition in the retail electricity market significantly depreciates the degree of cost pass-through. In competitive markets, firms face pressure to maintain competitive prices, causing them to absorb some of the increased costs or find alternative ways to manage costs.
- **Demand elasticity.** Demand elasticity refers to how sensitive consumers are to price changes. Consumers might not significantly reduce consumption even with price hikes for inelastic goods (like electricity, which is essential). This allows businesses to pass through more of the cost increase. Conversely, businesses might be less likely to raise prices significantly for elastic goods (where consumers can easily switch to alternatives). The more elastic supply is relative to demand, the greater the extent of pass-through; the greater the elasticity of demand relative to supply, the smaller the extent of pass-through. However, provided supply does not decrease as price increases and demand does not increase as price increases, pass-through will never exceed 100%. (6)
- **Global Energy markets.** Changes in energy costs in one locale can have a substantial effect on costs and supply elements in other parts of the world, influencing the pass-through behavior of businesses and governments. We recently observed this effect, initiated by the Russian-Ukrainian war.
- **Government Policies.** Policymakers might utilize price control mechanisms or offer subsidies to counteract the negative effects of electricity price hikes on consumer welfare. The European Union (EU) implemented stricter environmental regulations during the period I am focused on, including those targeting emissions from coal-fired power plants(IED, EU ETS). These regulations could lead to increased costs for power plants using fossil fuels, potentially pushing up electricity prices. The Czech Republic, as an EU member state, had to comply with these regulations.

One of the historical examples of the electricity cost pass-through effect is the California crisis of 2000-2001. Deregulation of California's electricity market coupled with drought and manipulation by utility companies has led to skyrocketing wholesale electricity prices. These costs were passed on to consumers, resulting in significantly higher prices for residential and commercial energy bills. The crisis underscored the vulnerability of consumers to

fluctuations in energy prices and highlighted the potential for market distortions to amplify the pass-through of costs, emphasizing the need for robust regulatory oversight and market transparency to safeguard against such disruptions.(14)

Another example of the pass-through effect happened in 2006 when Europe experienced an energy crisis characterized by a sharp increase in natural gas prices, driven primarily by supply shortages and geopolitical tensions. The surge in natural gas prices cascaded electricity prices across the continent, as many European countries rely heavily on natural gas for electricity generation. The pass-through of these increased costs resulted in higher energy bills for consumers and strained the finances of businesses reliant on electricity.

In the examples above, we see how changes in the prices of primary energy sources such as natural gas and coal are reflected in electricity prices and ultimately affect inflation. Once again, we are faced with a lack of information specifically focused on fluctuations in electricity prices.

2. Data

To understand the model's foundation, I will first describe sample variables. Most of them are taken from government sources - the Czech Statistical Office's (CZSO) Public Database and the Czech National Bank (ČNB) database. The dataset spanning from January 2015 to December 2019 encompasses monthly observations.

2.1 Dataset description

The dependent variable of interest **inflation**, is measured by the Consumer Price Index (CPI), the values of which are published by the CZSO in a public database. This index measures households' average price changes for a basket of consumer goods and services.

The work primarily focuses on whether short-term or long-term changes in the **price of electricity** are significantly associated with changes in the CPI. The *elec* indicator represents the market price of electricity in euros per megawatt-hour (MWh) reflecting dynamic shifts influenced by supply-demand dynamics and other market factors. The data is available on the <https://ember-climate.org/> website in annual, daily, and monthly format. Hourly data are obtained from ENTSO-e and cleaned, with missing values interpolated from nearby values. This data is then aggregated to produce balanced daily and monthly average values for each country.

The model incorporates **the exchange rate** (*exr*) to assess its impact on inflation, particularly concerning international trade and price transmission. The exchange rate pass-through effect puts heavy pressure on inflation targeting according to the balance sheet effect. The exchange rate depreciation can simultaneously increase the domestic firm cost and prices of foreign goods, which can also transform into domestic inflation. Moreover, the exchange rate pass-through effect suggests that exchange rate shocks can transfer global high commodity prices into domestic inflation (15). Monthly average exchange rates (CZK/EUR) are obtained from the ČNB website.

The unemployment rate (*unem*) can affect inflation through the Phillips curve. According to it, there is an inverse relationship between unemployment and inflation rates. This means that a falling unemployment rate usually leads to a higher inflation rate and vice versa. Data for this variable is retrieved from CZSO public publications.

Interest rate (*int*) is hypothesized to influence inflation dynamics through various channels. Changes in interest rates can impact aggregate demand, investment decisions, and inflation expectations, thereby exerting pressure on overall price levels in the economy. This work uses a PRIBOR rate, sourced from the Czech National Bank (ČNB) website.

Another explanatory variable selected is **industrial production index** (*ind*). It acts here as

an indicator of the overall performance of the industry, as it can reflect economic activity and demand, which can affect the price level and therefore also inflation. It was chosen due to its more frequent publication compared to GDP, typically used for economic growth assessment.

3. Methodology

Existing studies prevalently employ Vector Autoregression (VAR), Non-linear Autoregressive Distributed Lag (NARDL), Autoregressive Distributed Lag (ARDL), or ARDL-Error Correction Model (ARDL-ECM) methodologies. (1)(4) (15)

Informed by the previous research, I will develop a model with a structure similar to those that investigate the impact of oil prices on inflation. It is established, that the oil pass-through mechanism often shows asymmetric trends, however, I will attempt to catch the causal relationship using linear modeling.

The goal of this section is to describe the methods used in this paper and the concepts underlying them.

3.1 Stationarity

For robust estimation and explicable results, ARDL models require stationarity in both the dependent and independent variables. Stationarity implies that the variables exhibit a stable mean, variance, and covariance structure over time, ensuring the relationships captured by the model are consistent and explain meaningful relationships.

A strictly stationary time series is one for which the probabilistic behavior of every collection of values(16) $\{x_{t_1}, x_{t_2}, \dots, x_{t_k}\}$ is identical to that of the time-shifted set $\{x_{t_1+h}, x_{t_2+h}, \dots, x_{t_k+h}\}$ That is,

$$P(x_{t_1} \leq c_1, \dots, x_{t_k} \leq c_k) = P(x_{t_1+h} \leq c_1, \dots, x_{t_k+h} \leq c_k) \quad (3.1)$$

for all $k = 1, 2, \dots$,

all time points $t_1, t_2, \dots, t_k$,

all numbers $c_1, c_2, \dots, c_k$,

and all time shifts $h = 0, \pm 1, \pm 2, \dots$

Real-world data is barely achievable to call strictly stationarity. Henceforth, we will use the term stationary to mean weakly stationary.

A weak stationary process imposes conditions only on the first two moments of the series. This means the process has the same mean at all time points, and that the covariance between the values at any two-time points, t and $t-k$, depends only on k , the difference between the two times, and not on the location of the points along the time axis. In mathematical terms:

$$E(X_t) = \mu \quad \text{for all } t \quad (3.2)$$

$$E(X_t^2) = \sigma^2 \quad \text{for all } t \quad (3.3)$$

$$\text{cov}(X_t, X_k) = \text{cov}(X_{t+s}, X_{k+s}) \quad \text{for all } t, k, s \quad (3.4)$$

In other words, we mean flat-looking series, without trend, with constant variance over time and with no periodic fluctuations (seasonality) or autocorrelation.

3.2 Augmented Dickey-Fuller test

This test is an augmented version of a usual Dickey-Fuller test, which is used for testing the presence of unit root in AR(1) time series model. The test has been extended to handle unit root analysis in AR processes with a higher number of lags.

The presence of unit root essentially means a shock to the series has a permanent effect, and the series doesn't revert back to its mean over time.

The ADF test involves estimating an autoregressive (AR) regression of the following form:

$$\Delta Y_t = \alpha + \beta Y_{t-1} + \sum_i \gamma_i \Delta Y_{t-i} + \epsilon_t \quad (3.5)$$

The test-statistic formula is:

$$t_{\hat{\gamma}_i} = \frac{\hat{\gamma}_i}{SE(\hat{\gamma}_i)} \quad (3.6)$$

The test is conducted under the following assumptions:

- Null Hypothesis (H0): There exists a unit root in the time series and it is non-stationary
 $\beta = 0$
- Alternate Hypothesis (H1): There is no unit root in the time series and it is stationary
 $\beta < 0$

3.3 Autocorrelation

In time series analysis autocorrelation refers to the correlation between a series and a lagged version of itself. It can also be referred to as lagged correlation or serial correlation.

As in cross-sectional analysis, working with time series data requires an assumption on the absence of dependency between error terms, in this case, the errors in 2 different periods must be uncorrelated. Mathematically:

$$\text{Corr}(u_t, u_s | X) \neq 0, \text{ for all } t \neq s \quad (3.7)$$

If residuals in an ARDL model exhibit autocorrelation, it breaks key assumptions of the model. This can lead to inaccurate estimates of the model's parameters and unreliable conclusions about the relationships between variables.

3.4 ACF and PACF function

Autocorrelation plots are a crucial step in evaluating the suitability of a chosen model. By computing autocorrelations of data points at sequential time intervals (lags), we can assess the presence of autocorrelation, which would indicate a lack of randomness. (17)

Partial autocorrelation plots are a commonly used tool for model identification in Box-Jenkins models. The partial autocorrelation at lag k is the autocorrelation between X_t and X_{t-k} that is not accounted for by lags 1 through $k - 1$.

Respectively, partial autocorrelation is useful in identifying the order of an autoregressive model. The partial autocorrelation of an $AR(p)$ process is zero at lag $p + 1$ and greater. If the sample autocorrelation plot indicates that an AR model may be appropriate, then the sample partial autocorrelation plot is examined to help identify the order.(18)

3.5 Autoregressive Distributed Lag model

This thesis adopts the Autoregressive Distributed Lag (ARDL) approach to cointegration originally formulated by Pesaran (19) and later refined by Pesaran, Shin, and Smith (20) with the bounds testing approach. ARDL was chosen due to its absence of requirements on variables to be integrated at first difference order, $I(1)$, and mandatory equal lag length in the model.

The ARDL model has a general form where y , modeled in levels or differences, is a function of itself (in lagged levels or differences), up to k variables x , either in simultaneous levels, lagged levels, contemporaneous differences, or lagged differences. It is a logical extension of $AR(p)$ model.

General ARDL (p,q) model is given by:

$$\psi(L)y_t = \delta + \theta(L)x_t + \epsilon_t \quad (3.8)$$

where $\psi(L)$ is an order p polynomial that, for stability, has to be stationary, and $\theta(L)$ is an order q polynomial. Expanding the lag polynomials, equation (3.8) can be written as

$$y_t = \delta + \psi_1 y_{t-1} + \dots + \psi_p y_{t-p} + \theta_0 x_t + \theta_1 x_{t-1} + \dots + \theta_q x_{t-q} + \epsilon_t \quad (3.9)$$

In this equation, y_t represents the dependent variable, δ is the constant term, $\psi_1 \dots \psi_p$ - coefficient of autoregressive term, $\theta_0 \dots \theta_q$ are dynamic marginal effects of explanatory variables on y_t , and ϵ_t is a disturbance.

The model has standard assumptions regarding residuals:

- $Cov(\epsilon_t, \epsilon_s) = 0$, for $t \neq s$
- $Var(\epsilon_t) = \sigma^2$

As long as it can be assumed, that the error terms are stationary, the ARDL model can be consistently estimated by Ordinary least squares(21).

There are a lot of advantages to using this particular modeling technique. A distinct advantage of the ARDL approach lies in its ability to handle variables integrated at different orders (19) which solves the problem of non-stationary time series data. ARDL also allows treating serial correlation problems (22) by lengthening the lags.

4. Empirical analysis

All analysis was performed using the R programming language. For creating ARDL models "dynlm" package is used. "Dynlm" was designed specifically for working with dynamic linear models. Another advantage is the ability to exclude some lags, still capturing the effect of another.(23)

The presentation of the results is made using the "stargazer" package, which creates LATEX code, HTML code and ASCII text for well-formatted regression tables, with multiple models side-by-side, as well as for summary statistics tables, data frames, vectors and matrices.(24)

4.1 Testing for the stationarity

A big difference between minimum and maximum values through the majority of the series indicates the data's non-stationarity. However, this is only an assumption that additional tests should confirm. This problem can be resolved with data transformation, enhancing overall model performance.

Specifically, transforming variables to their logarithmic form is considered. This approach allows for interpreting changes and their impact on inflation as percentages. In addition, this transformation will help to stabilize the variance, improving the accuracy of the model's predictions. After modification, I mark the series as *lcpi*, *lelec* etc.

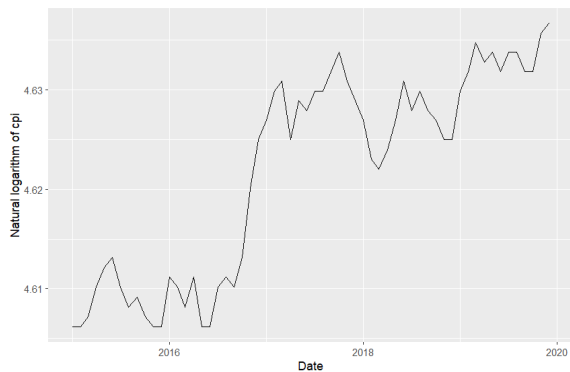


Figure 4.1: *lcpi*.

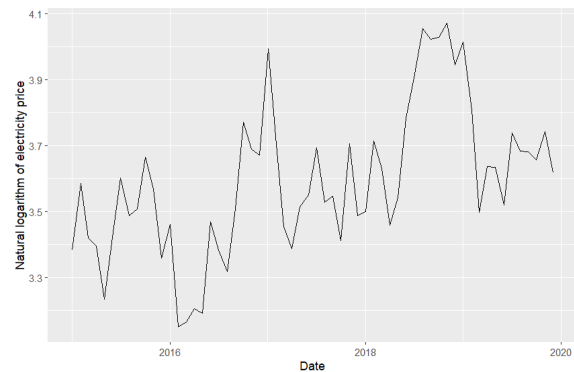


Figure 4.2: *lelec*

Visual inspection of the data suggests non-stationarity in all variables (viz fig 4.1 - 4.6). The results of the Augmented Dickey-Fuller (ADF) tests supported the presence of a unit root (non-stationarity) in all variables at all significance levels, except for the unemployment rate. (Table 4.1)

To address this issue, first-differencing is employed as a transformation technique. This involves calculating the difference between consecutive observations in the time series. (After differencing the data I will refer to them as *dlcpi*, *dlelec* etc. where *d* stands for the first dif-

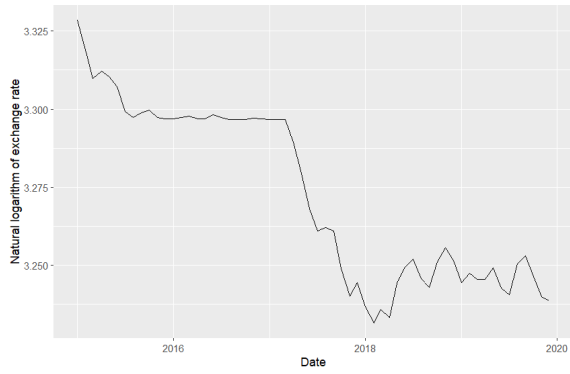


Figure 4.3: *lexr*

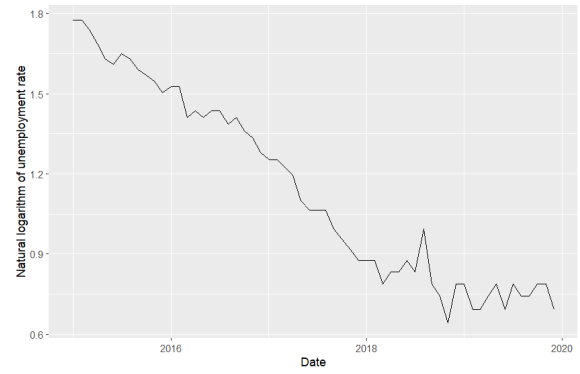


Figure 4.4: *lunem*

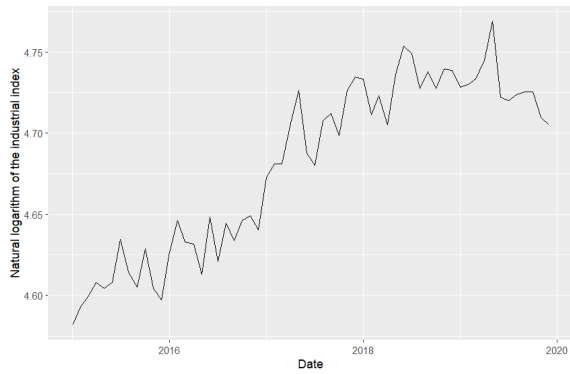


Figure 4.5: *lind*

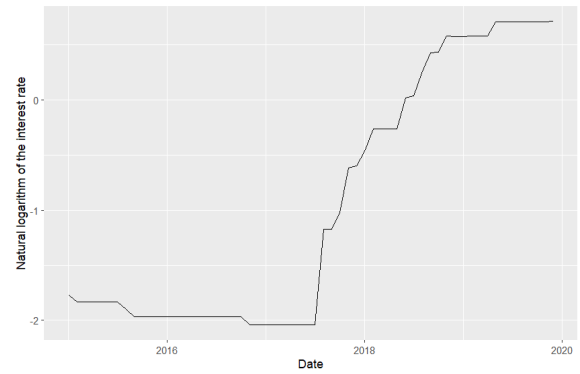


Figure 4.6: *lint*

Table 4.1: ADF test results for logarithmic variables(without drift and trend)

<i>Variable</i>	<i>Lag length</i>		
	1	2	3
lcpi	1.345	1.266	1.141
lelec	-0.061	0.124	0.108
lexr	-1.519	-1.889	-1.512
lunem	-1.512	-3.541***	-4.22***
lind	1.078	1.435	1.604
lint	-1.202	-1.155	-1.242

Note: *p<0.1; **p<0.05; ***p<0.01

ference.) Subsequent ADF tests on the differenced data confirm stationarity for all variables (see Table 4.2). With this accomplished the paper paves the way to further analysis.

The summary statistics presented in Table 4.3 suggest that the transformed and differenced data exhibits relatively stable means and moderate variability for most variables, except for *dlint*, which shows a wider range.

Table 4.2: ADF test results for logarithmic variables after differencing (without drift and trend)

Variable	Lag length		
	1	2	3
dlcpi	-4.721***	-3.947***	-3.770***
dlelec	-6.985***	-4.676***	-3.948***
dlexr	-5.772***	-3.349**	-2.745**
dlunem	-5.427***	-4.835***	-3.071**
dlind	-7.734***	-5.974***	-4.254***
dlint	-3.926***	-2.179*	-2.065*
Note:	*p<0.1; **p<0.05; ***p<0.01		

Table 4.3: Summary statistics after logarithmic transformation and differencing

Statistic	Mean	St. Dev.	Min	Max
dlcpi	0.001	0.003	-0.006	0.007
dlelec	0.004	0.160	-0.311	0.323
dlexr	-0.002	0.005	-0.012	0.011
dlunem	-0.018	0.059	-0.205	0.160
dlind	0.002	0.018	-0.047	0.035
dlint	0.042	0.138	-0.074	0.869

4.2 Model specification

First, the simplest model that explicit the relationship between the variable of interest, its first lag, and electricity prices (ARDL(1,0)). The results are tabulated in Table 4.4 establishing no association between electricity prices and inflation. Very low adjusted R2 shows that electricity prices have little to no power to explain the inflation variance.

Nevertheless, we only found out that the current model configurations are not sufficient to reach a conclusive judgment. Revising specifications or additional data might be necessary.

For that reason, I added more independent variables and now my model is mathematically specified as:

$$\begin{aligned}
 \Delta lcpit = \beta + \sum_{j=1}^p \gamma_j \Delta lcpit-j + \sum_{i=1}^{q_1} \beta_1 \Delta lelect_{t-i} + \sum_{i=1}^{q_2} \beta_2 \Delta lexr_{t-i} + \sum_{i=1}^{q_3} \beta_3 \Delta lunem_{t-i} + \\
 + \sum_{i=1}^{q_4} \beta_4 \Delta lind_{t-i} + \sum_{i=1}^{q_5} \beta_5 \Delta lint_{t-i} + \epsilon_t
 \end{aligned} \tag{4.1}$$

Hence, the following task involves the identification of parameters p and $q_{1,2..5}$ for the model performance.

Table 4.4: ARDL(1,0) model results

	<i>Dependent variable:</i>
	lcpi
L(dlcpi, 1)	0.081 (0.136)
dlelec	−0.0003 (0.002)
Constant	0.0005 (0.0004)
Observations	58
R ²	0.007
Adjusted R ²	−0.029
Residual Std. Error	0.003 (df = 55)
F Statistic	0.206 (df = 2; 55)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

There is no universally agreed-upon method to determine appropriate p and q parameters for the ARDL model. Nevertheless, any approach should include theoretical consideration, using ACF and PACF functions and statistical criteria like the Akaike Information Criterion (AIC).

- Analyzing both ACF and PACF plots jointly allows us to pinpoint the most suitable lag structure for capturing the temporal dynamics within the data. While significant spikes in the ACF plot indicate potential lag relationships, the PACF isolates the unique effect of each lag. By rule of thumb, we compute ACF up to one-third to one-quarter of the length of the time series. (**Nkoro2016**)
- Statistical criteria can be used to select the lag length that minimizes the model's complexity while capturing the relevant information in the data. These criteria penalize models with too many lags, helping to avoid overfitting.
- Economic theory is the foundation for both variable selection and lag choice. Chapter 2 details the rationale behind the specific parameters used.

ACF and PACF functions for the first 3 variables don't bring enough information to consider lag length, however, analyzing plots for *dunem*, *dlind* and *dlint* I can spot spikes or further investigation (Fig. 4.7, Fig. 4.8)

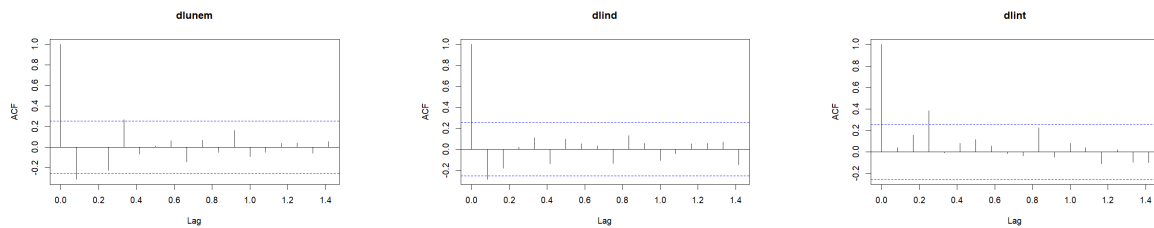


Figure 4.7: ACF correlograms for chosen variables

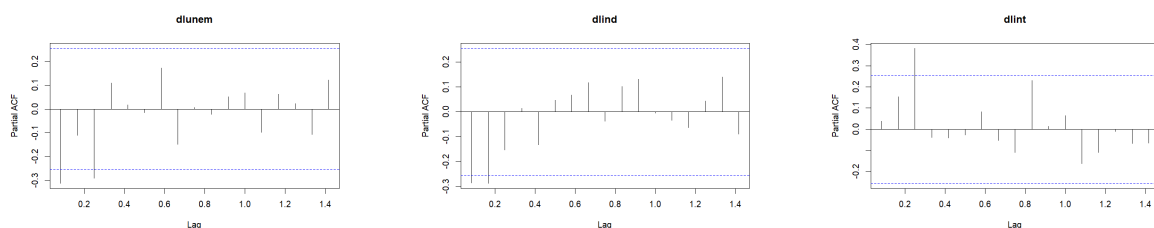


Figure 4.8: PACF correlograms for chosen variables

Analyzing those correlograms I decided, to substitute the values q_3, q_4 and q_5 with 3,2,3 respectively. The estimation now detected the 3rd lag of *dlint* as significant for the model, indicating a step in the right direction. (Table 4.5)

Investigating the autocorrelation function of residuals from the model ADRL(1,1,3,2,3), I spotted significant spikes at the 12th and 6th lags (Fig. 4.9) (since the data is presented monthly, the value 1 on the x-axis here means a year) This violates the model assumption and suggest that there are still unexplained relationships in the error term.

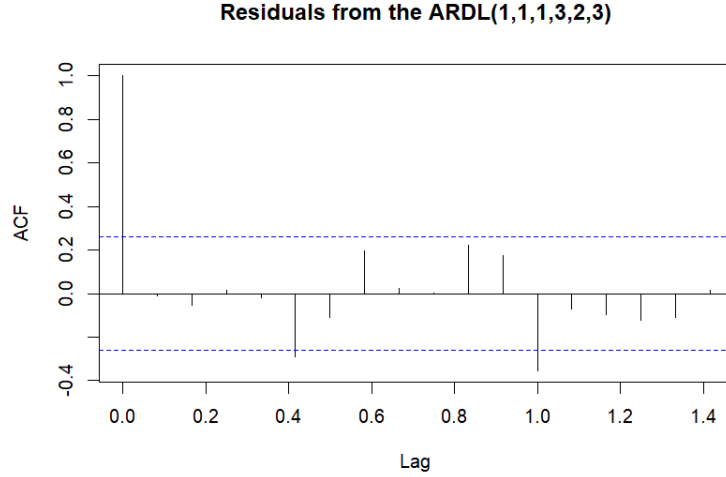


Figure 4.9: ACF correlogram

Considering the ACF correlogram I included those lags to the model, excluding effects from the previous delays. x_{t-2} is not usually omitted as a regressor even if its coefficient is not statistically significant. The only common exception to that rule is with seasonal data, where we might augment a short lag of 3 or 4 periods with a lag-12 coefficient when using monthly data to capture the effect of the same month in the preceding year, which is the case for my study.(17) It is also logical to assume that the period of a whole year can be relevant in such an analysis. It also helps to keep up with the model testimony, especially considering the small data sample. Results of ARDL((6,12),1,3,2,3) are presented in Table 4.5.

With the last modification, the explained variance presented by adjusted R^2 increases to around 21,26%. While the model still doesn't account for all factors, it's a step in the right direction.

This model already meets the assumptions on the absence of autocorrelation. This can be seen in Figure 4.10, furthermore, the lack of serial correlation up to 12 lag was confirmed by the Breusch-Godfrey test. According to the Breusch-Pagan test, the assumption of homoscedasticity is also met, since the test didn't show enough evidence to reject the null hypothesis. In simple terms, the variance and mean remain constant throughout the model.

Despite the facts above, I still cannot confirm the conjecture that electricity prices are impacting inflation. Furthermore, my residuals still don't look like white noise. My next modification includes checking the primary influence of the electricity prices by setting the lag length to 0 and simultaneously exploring the marginal effects of up to 6 lags.

Results (Table 4.4) from this estimation now capture the significance of both the current level (0th lag) and 6th lag coefficient of the electricity price *lelec* variable. As a result, the model's adjusted R^2 rises to 26.1% The p-value associated with this F-statistic is 0.02982. Since the p-value is less than the conventional significance level of 0.05, we can conclude that the overall model is statistically significant at the 5% level.

Table 4.5: ARDL(1,1,1,3,2,3) and ARDL((6,12),1,1,3,2,3) results

	<i>Dependent variable:</i>	
	lcpir	
	(1)	(2)
L(dlcpi, 1)	0.086 (0.137)	
L(dlcpi, 6)		−0.278* (0.143)
L(dlcpi, 12)		−0.531*** (0.151)
L(dlelec, 1)	−0.0002 (0.002)	0.002 (0.002)
L(dlexr, 1)	−0.017 (0.081)	−0.107 (0.078)
L(dlunem, 3)	0.0005 (0.006)	−0.002 (0.006)
L(dlind, 2)	−0.006 (0.021)	0.002 (0.021)
L(dlint, 3)	−0.005* (0.003)	−0.002 (0.003)
Constant	0.001 (0.0004)	0.001** (0.0004)
Observations	56	47
R ²	0.083	0.332
Adjusted R ²	−0.029	0.213
Residual Std. Error	0.003 (df = 49)	0.002 (df = 39)
F Statistic	0.744 (df = 6; 49)	2.774** (df = 7; 39)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 4.6: ARDL models results

	<i>Dependent variable:</i>	
	lcp <i>i</i>	
	(1)	(2)
L(dlcpi, 6)	−0.385** (0.150)	−0.349** (0.140)
L(dlcpi, 12)	−0.599*** (0.160)	−0.602*** (0.148)
L(dlelec, 0)	0.005* (0.003)	0.004* (0.002)
L(dlelec, 1)	0.002 (0.002)	
L(dlelec, 2)	0.002 (0.002)	
L(dlelec, 3)	−0.00003 (0.002)	
L(dlelec, 4)	0.003 (0.002)	
L(dlelec, 5)	−0.001 (0.003)	
L(dlelec, 6)	0.004* (0.003)	0.004* (0.002)
L(dlexr, 1)	−0.127 (0.079)	−0.117 (0.074)
L(dlunem, 3)	−0.003 (0.006)	−0.003 (0.006)
L(dlind, 2)	−0.016 (0.025)	−0.026 (0.021)
L(dlint, 3)	−0.004 (0.003)	−0.004 (0.003)
Constant	0.001** (0.0004)	0.001** (0.0004)
Observations	47	47
R ²	0.470	0.412
Adjusted R ²	0.261	0.288
Residual Std. Error	0.002 (df = 33)	0.002 (df = 38)
F Statistic	2.251** (df = 13; 33)	3.324*** (df = 8; 38)

Note:

*p<0.1; **p<0.05; ***p<0.01

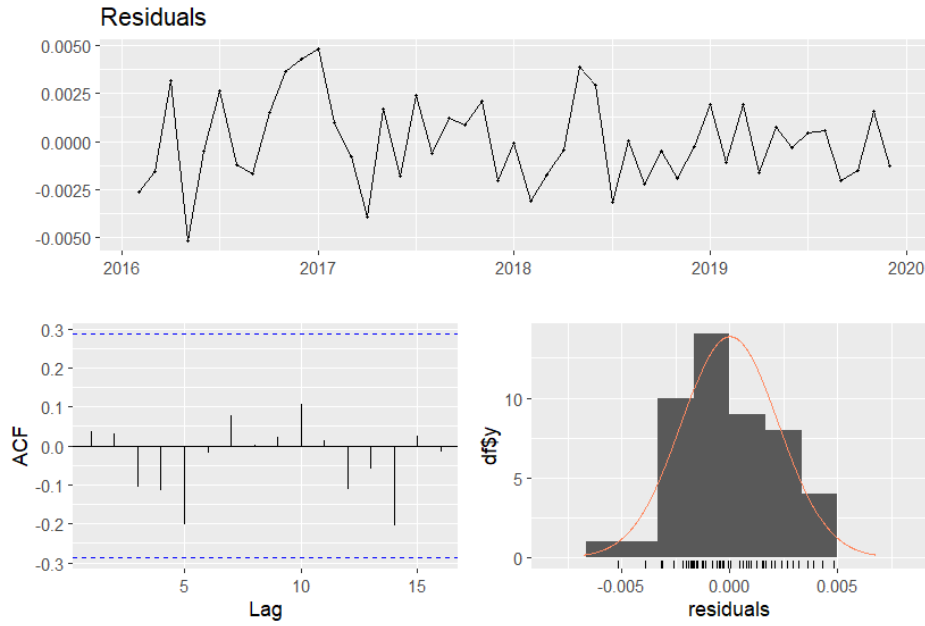


Figure 4.10: Plot, ACF correlogram and histogram of residuals from ADRL((12,6),1,1,3,2,3)

Again, after conducting diagnostic tests and visual inspection of the model's residuals I can verify its adherence to consistency criteria. (Figure 4.11)

Figure 4.12 presents diagnostic residual plots of the model with included up to 6th lags of *dlelec*. Based on the 1st "residuals vs Fitted" I can conclude, that there is not any non-linear pattern in unexplained variation. QQ plot indicates a normal distribution, subsequent Shapiro test confirms that. Scale-location plot also exhibits random spreading of the disturbances.

Lastly, I want to exclude the effect of the insignificant *dlelec* lags from 1 to 5 and test if my model would have more predictive power.

From the result presented in Table 4.4, I can still conclude that the coefficients before *dlelec* from the current period and its 6th lag are significant on the 10% confidence level. Furthermore, excluding parameters helped me receive higher adjusted R^2 .

Breusch-Godfrey test for serial correlation didn't find evidence of order up to 12. According to the Breusch-Pagan test, there is insufficient evidence to conclude that heteroscedasticity is present in the regression model. Diagnostic plots in Figure 4.13 also look appropriate for this model specification. ADF test of the residuals didn't find enough evidence of non-stationarity.

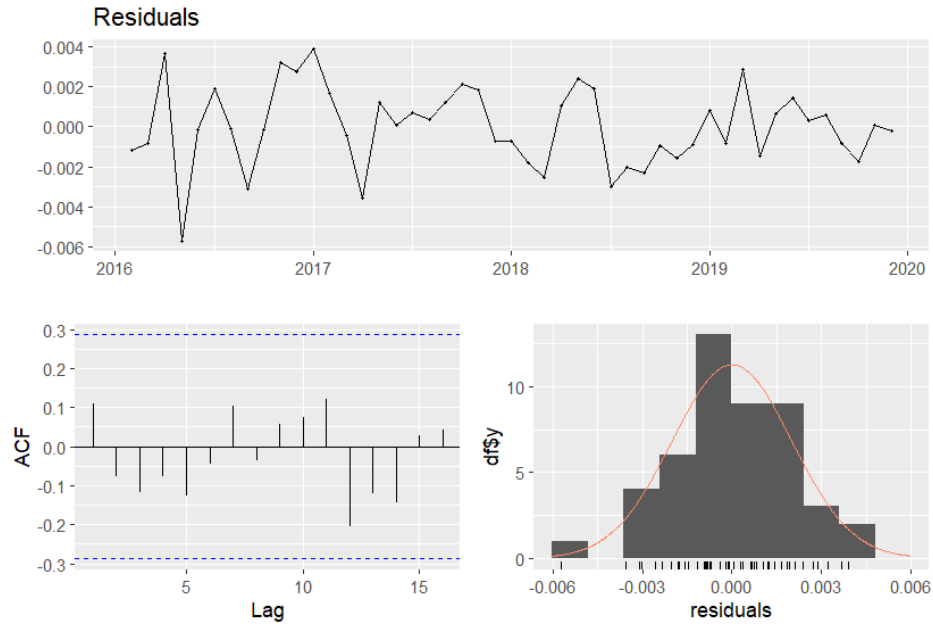


Figure 4.11: Plot, ACF correlogram and histogram of residuals from ADRL((12,6),1,1,3,2,3)

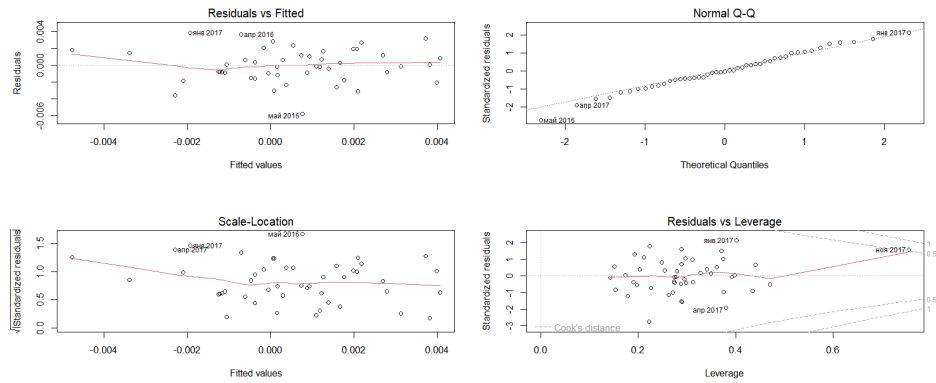


Figure 4.12: Diagnostic plots from ARDL((6,12),0:6,1,3,2,3) model

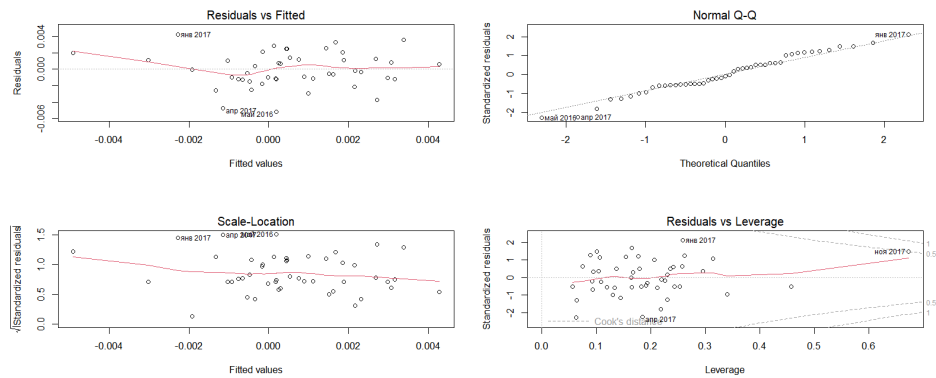


Figure 4.13: Diagnostic plots from ARDL((6,12),(0,6),1,3,2,3) model

Discussion

The biggest limitation in my analysis, except for personal lack of deep knowledge, was a small sample size of 60 monthly observations. A larger dataset could potentially yield more robust results regarding the electricity price effect.

Consequently, exploring cointegration and potentially expanding the model to its Error Correction Model (ECM) or Unrestricted Error Correction Model (UECM) form would be valuable next steps. This approach would allow for a more comprehensive examination of the long-run and short-run dynamics within the model, potentially mitigating the influence of the limited sample size.

Also, I wanted to explain the misalignment between the abstract, where I stated that "Results showed that both types of relationships are present between these variables" and the final results. A mistake was made in the initial stage of my work, which was discovered during the process of writing this paper.

Conclusion

This paper aimed to find a relationship and quantitative assessment between electricity price fluctuations and changes in the inflation rate in the Czech Republic.

The analysis utilized monthly data from January 2015 to December 2019 and incorporated lags of the dependent variable (*lcpi*), electricity prices (*lelec*), exchange rate (*lexr*), unemployment rate (*lunem*), interest rate (*lint*), and industrial production rate (*lind*).

The key findings of the analysis are:

- Both the 6-month and 12-month lagged values of charging in CPI have statistically significant negative coefficients, indicating that changes in past values of the consumer price index negatively impact its current level.
- None of the coefficients for the independent variables (*lelec*, *lexr*, *lunem*, *lind*, *lint*) are statistically significant, suggesting that within the chosen model specification, they might not have a strong short-run impact on *lcpi*.
- The overall model fit is low, explaining around 28,8% of the variance in *lcpi* after adjusting for the number of regressors.

While the final model shows progress in addressing autocorrelation and overall significance, the impact of electricity prices on inflation remains unclear. It is important to acknowledge that the lack of statistical significance for some coefficients doesn't necessarily imply the absence of any effect. Further exploration using alternative specifications, additional variables, or data transformations could be necessary to strengthen the model and confirm the electricity price effect.

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